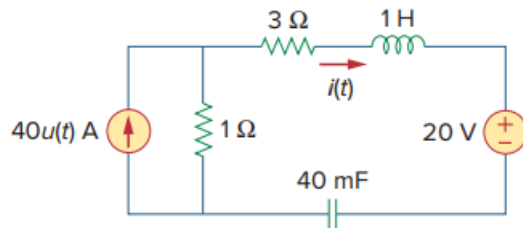


Problem

For the network in Fig. 8.89, find $i(t)$ for $t > 0$.

Figure 8.89:



Step-by-step solution

Step 1 of 13

Refer to Figure 8.89 in the text book.

At $t < 0$, the capacitor acts as open circuit and inductor acts as short circuit. The value of the current source is 0 A since $u(t)$ is 0 at $t < 0$.

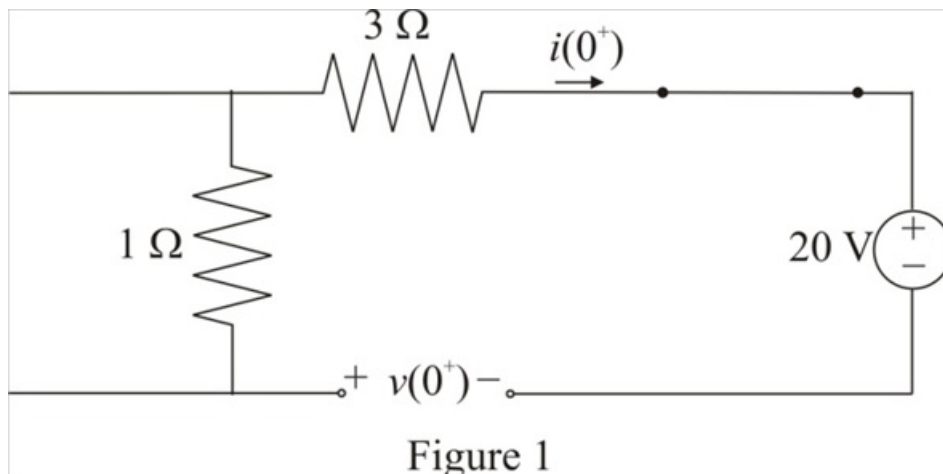


Figure 1

The value of the current $i(0^-)$ is 0 A .

Step 2 of 13

Calculate the value of the voltage $v(0^-)$.

$$\begin{aligned} v(0^-) &= 20 + (3 + 1)i(0^-) \\ &= 20 + 4i(0^-) \end{aligned}$$

Substitute 0 A for $i(0^-)$.

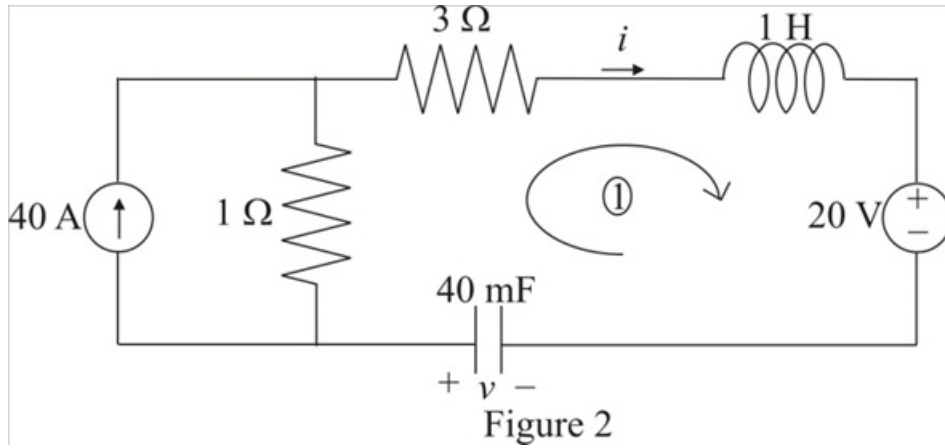
$$\begin{aligned} v(0^-) &= 20 + (4)(0) \\ &= 20 + 0 \\ &= 20 \text{ V} \end{aligned}$$

Therefore, the value of the voltage $v(0^-)$ is 20 V .

Step 3 of 13

Draw the circuit diagram at $t > 0$.

The value of the current source is 40 A since $u(t)$ is 1 at $t > 0$.



Step 4 of 13

The voltage across the capacitor and the current through the inductor will not change instantaneously. Hence, at $t > 0$ the value of the voltage across the capacitor $v(0^+)$ is 20 V and the value of the current $i(0^+)$ is 0 A .

Notice that the voltage across the capacitor and the current through the inductor are continuous. Therefore, the values at $t = 0$ are $v(0) = 20\text{ V}$ and $i(0) = 0\text{ A}$.

Step 5 of 13

Apply Kirchhoff's Voltage Law in loop-1.

$$1(i - 40) + 3i + 1 \frac{di}{dt} + 20 - v = 0$$

$$i - 40 + 3i + 1 \frac{di}{dt} + 20 - v = 0$$

$$\frac{di}{dt} - 20 + 4i - v = 0$$

$$\frac{di}{dt} = v + 20 - 4i$$

Calculate the value of $\frac{di}{dt}$ for $t = 0$.

$$\frac{di(0)}{dt} = v(0) + 20 - 4i(0)$$

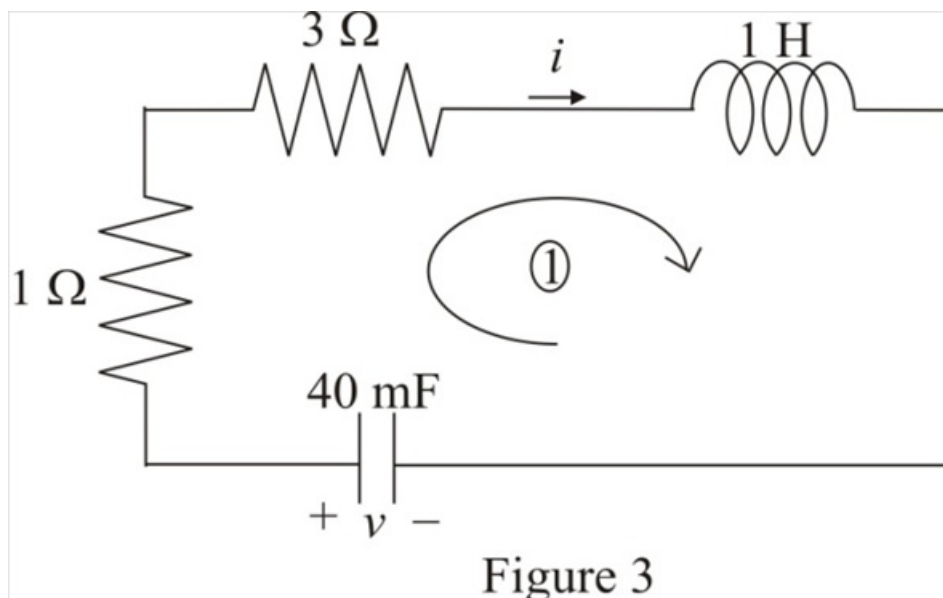
Substitute 20 V for $v(0)$ and 0 A for $i(0)$.

$$\begin{aligned} \frac{di(0)}{dt} &= 20 + 20 - 0 \\ &= 40\text{ A/s} \end{aligned}$$

Therefore, the value of $\frac{di(0)}{dt}$ is 40 A/s .

Step 6 of 13

Deactivate the sources to estimate the transient response of the circuit.



Step 7 of 13

Apply Kirchhoff's Voltage Law in loop-1.

$$1i + 3i + 1 \frac{di}{dt} + \frac{1}{40 \text{ m}} \int i \, dt = 0$$

$$4i + \frac{di}{dt} + \frac{1}{40 \text{ m}} \int i \, dt = 0$$

Apply derivative with respect to t .

$$4 \frac{di}{dt} + \frac{d^2i}{dt^2} + 25i = 0$$

$$\frac{d^2i}{dt^2} + 4 \frac{di}{dt} + 25i = 0$$

Treat $\frac{d}{dt}$ as s .

$$s^2i + 4si + 25i = 0$$

$$(s^2 + 4s + 25)i = 0$$

[Comments \(1\)](#)



Anonymous

Why is the voltage drop across the capacitor considered positive if positive current is entering the negative terminal?

Step 8 of 13

Consider the characteristic equation of the circuit.

$$s^2 + 4s + 25 = 0$$

$$(s + 2 + j4.583)(s + 2 - j4.583) = 0$$

The roots of the characteristic equation are $s = -2 \pm j4.583$.

The circuit exhibits under-damped response since the roots of the characteristic equation are complex.

Compare the roots to $s = \alpha \pm j\omega_d$.

The values of α and ω_d are -2 Np/s and 4.583 rad/s respectively.

Step 9 of 13

Draw the circuit for $t = \infty$.

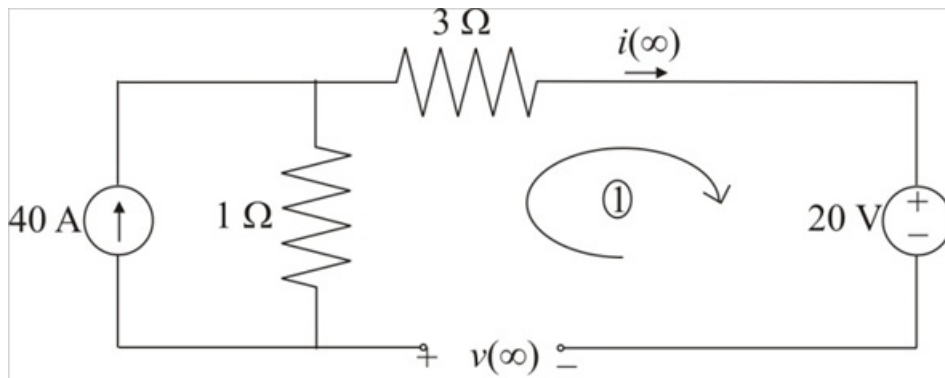


Figure 4

The value of the current at $t = \infty$ is 0 A.

Step 10 of 13

Write the general solution for the current $i(t)$ for under-damped condition.

$$i(t) = i(\infty) + (A_1 \cos \omega_d t + A_2 \sin \omega_d t) e^{\alpha t}$$

Substitute 0 A for $i(\infty)$, 4.583 rad/s for ω_d and -2 Np/s for α .

$$i(t) = 0 + (A_1 \cos 0.6t + A_2 \sin 0.6t) e^{-2t} \quad i(t) = (A_1 \cos 4.583t + A_2 \sin 4.583t) e^{-2t} \dots (1)$$

Step 11 of 13

Recall equation-1.

$$i(t) = (A_1 \cos 4.583t + A_2 \sin 4.583t) e^{-2t}$$

Substitute 0 for t .

$$\begin{aligned} i(0) &= (A_1 \cos 4.583(0) + A_2 \sin 4.583(0)) e^{-2(0)} \\ &= (A_1 \cos 0 + A_2 \sin 0) e^0 \\ &= (A_1(1) + A_2(0))(1) \\ &= A_1 + 0 \end{aligned}$$

Substitute 0 A for $i(0)$.

$$A_1 = 0$$

Therefore, the value of the constant A_1 is 0.

Step 12 of 13

Recall equation (1).

$$i(t) = (A_1 \cos 4.583t + A_2 \sin 4.583t)e^{-2t}$$

Apply derivative with respect the t .

$$\begin{aligned} \frac{di(t)}{dt} &= \frac{d}{dt}[(A_1 \cos 4.583t + A_2 \sin 4.583t)e^{-2t}] \\ &= (A_1 \cos 4.583t + A_2 \sin 4.583t) \frac{d}{dt}e^{-2t} + e^{-2t} \frac{d}{dt}(A_1 \cos 4.583t + A_2 \sin 4.583t) \\ &= \left[-2e^{-2t}(A_1 \cos 4.583t + A_2 \sin 4.583t) + \right. \\ &\quad \left. e^{-2t}(-4.583A_1 \sin 4.583t + 4.583A_2 \cos 4.583t) \right] \end{aligned}$$

Substitute 0 for t .

$$\begin{aligned} \frac{di(0)}{dt} &= \left\{ \begin{aligned} &-2e^{-2(0)}(A_1 \cos 4.583(0) + A_2 \sin 4.583(0)) \\ &+ e^{-2(0)}(-4.583A_1 \sin 4.583(0) + 4.583A_2 \cos 4.583(0)) \end{aligned} \right\} \\ &= -2e^0(A_1 \cos(0) + A_2 \sin(0)) + e^0(-4.583A_1 \sin(0) + 4.583A_2 \cos(0)) \\ &= -2(1)(A_1(1) + A_2(0)) + (1)(-4.583A_1(0) + 4.583A_2(1)) \\ &= -4.583A_1 + 4.583A_2 \end{aligned}$$

Substitute 0 for A_1 and 40 A/s for $\frac{di(0)}{dt}$.

$$40 = -2(0) + 4.583A_2$$

$$40 = 0 + 4.583A_2$$

$$4.583A_2 = 40$$

$$A_2 = \frac{40}{4.583}$$

$$A_2 = 8.7$$

Therefore, the value of the constant A_2 is 8.7.

Step 13 of 13

Recall equation (1).

$$i(t) = (A_1 \cos 4.583t + A_2 \sin 4.583t)e^{-2t}$$

Substitute 8.7 for A_2 and 0 for A_1 .

$$\begin{aligned} i(t) &= ((0) \cos 4.583t + 8.7 \sin 4.583t)e^{-2t} \\ &= (0 + 8.7 \sin 4.583t)e^{-2t} \\ &= 8.7e^{-2t} \sin 4.583t \text{ A} \end{aligned}$$

Therefore, the value of the current $i(t)$ is $8.7e^{-2t} \sin 4.583t \text{ A}$.